

Tutorial Sheet - V

Q.1. Let A be an $n \times n$ matrix over a $\mathbb{F}$.

For any $\alpha \in \mathbb{F}$, show that $\det(\alpha A) = \alpha^n \det(A)$.

Q2. Let A be a 2×2 real matrix be such that $A^2 = 0$. Show that $\det(cI - A) = c^2$ for any real number c .

Q3. An $n \times n$ matrix A over $\mathbb{F}$ is skew-symmetric if $A^T = -A$. If A is a skew-symmetric $n \times n$ matrix with complex entries and n is odd, show that $\det(A) = 0$.

Q4. An $n \times n$ matrix over $\mathbb{F}$ is orthogonal if $AA^T = I$. If A is orthogonal show that $\det(A) = \pm 1$.
Give an example of an orthogonal matrix with $\det(A) = -1$.

Q.5. Let S_n be the permutation group of n -symbols. Show that $\text{Sgn} : S_n \rightarrow \{\pm 1\}$ defined by $\sigma \mapsto \text{Sgn}(\sigma)$ is a group homomorphism, that is $\text{Sgn}(\sigma_1 \circ \sigma_2) = \text{Sgn}(\sigma_1) \cdot \text{Sgn}(\sigma_2)$, $\forall \sigma_1, \sigma_2 \in S_n$.

Q.6. Find an eigenvalue of the matrix

$$\begin{bmatrix} 2022 & 2 & 3 \\ 2 & 2025 & 6 \\ 3 & 6 & 2030 \end{bmatrix}.$$

Q7. Let $T: \mathbb{F}^n \rightarrow \mathbb{F}^n$ be a linear map.

Show that

$$\det(T(\alpha_1), \dots, T(\alpha_n)) = c \det(\alpha_1, \dots, \alpha_n).$$

for all $\alpha_1, \dots, \alpha_n \in \mathbb{F}^n$, where

$c = \det(T(e_1), \dots, T(e_n))$, and $e_1, \dots, e_n$ is the standard basis for $\mathbb{F}^n$.

Q8. Find matrices $A, B \in M_{n \times n}(\mathbb{C})$ and scalars $\lambda, \beta \in \mathbb{C}$ such that

(i) λ is an eigenvalue of A , β is an eigenvalue of B but $\lambda + \beta$ is not an eigenvalue of $A + B$.

(ii) $\lambda\beta$ is not an eigenvalue of AB .

Q9. Find determinant of the matrix

$$\begin{bmatrix} 7 & 2 & 2 & 2 & 2 \\ 2 & 7 & 2 & 2 & 2 \\ 2 & 2 & 7 & 2 & 2 \\ 2 & 2 & 2 & 7 & 2 \\ 2 & 2 & 2 & 2 & 7 \end{bmatrix}.$$